

Chapter 41: Gamma Matrix Technology

March 29, 2026

Ch.41 is the reference manual for all γ -matrix algebra. Every equation in Ch.39, 40, and 43 uses these identities.

1 Weyl (Chiral) Basis

In the chiral (Weyl) basis:

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -\sigma^i \\ +\sigma^i & 0 \end{pmatrix}, \quad (1)$$

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -I & 0 \\ 0 & +I \end{pmatrix}. \quad (2)$$

Chiral projectors: $P_L = (1 - \gamma^5)/2$, $P_R = (1 + \gamma^5)/2$.

2 Clifford Algebra

$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}_4$. Consequences: $(\gamma^0)^2 = +\mathbf{1}$, $(\gamma^i)^2 = -\mathbf{1}$.

3 Trace Theorems (Complete Set)

$$\text{Tr}[\mathbf{1}] = 4, \quad \text{Tr}[\gamma^\mu\gamma^\nu] = 4\eta^{\mu\nu}, \quad (3)$$

$$\text{Tr}[\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}). \quad (4)$$

4 Chiral Anomaly (Adler-Bell-Jackiw)

$$\partial_\mu j^{5\mu} = \frac{e^2}{16\pi^2} \text{Tr}[\gamma^5 F_{\mu\nu} F^{\mu\nu}] = \frac{e^2}{8\pi^2} F_{\mu\nu} \tilde{F}^{\mu\nu}.$$

This is an exact quantum anomaly — not a symmetry of the Lagrangian but a consequence of regularization.

5 Fierz Identities

$$[\bar{\psi}_A \gamma^\mu (1 \pm \gamma^5) \psi_B][\bar{\psi}_C \gamma_\mu (1 \mp \gamma^5) \psi_D] = [\bar{\psi}_A \gamma^\mu (1 \pm \gamma^5) \psi_D][\bar{\psi}_C \gamma_\mu (1 \mp \gamma^5) \psi_B] .$$

Used in: Fermi theory, CKM matrix elements, SUSY superpotential matching.

6 γ^5 and the Chiral Basis

In the chiral basis, the 4-component notation splits cleanly: $\gamma^\mu = (0, \sigma^\mu; \bar{\sigma}^\mu, 0)$. The connection to 2-component Weyl spinors is: $u_+(k) \rightarrow \lambda_\alpha$, $u_-(k) \rightarrow \tilde{\lambda}_{\dot{\alpha}}$. This is the bridge to the spinor-helicity formalism of Ch.42.